

Shivani Kalamadi

@ skalamadi@ucdavis.edu |  linkedin.com/in/shivanikalamadi |  shivani-k |  shivanikalamadi.com |  San Mateo, CA

EDUCATION

University of California, Davis

MS and PhD. in Computer Science; **GPA: 3.88/4.00**

Davis, CA

Sep 2022 – Present

SDM College of Engineering and Technology

B.E. in Computer Science; **GPA: 9.00/10.00**

Karnataka, India

Aug 2015 – Jun 2019

RESEARCH

Internet Measurement

UC Davis

Advisor: Prof. A. Gamero-Garrido – *ImPLYLab*

Jun 2024 – Present

Digital Divide & Internet Access

- Investigated U.S. internet access disparities through a socio-economic lens across six metropolitan regions, using five years (2021–2025) of **Ookla** crowd-sourced speed tests at the census block group level and stratifying urban areas by population density to surface trends hidden by aggregate urban/rural comparisons.
- Identified and corrected for sampling bias, and used a Random Forest regression model to correlate socio-economic features with network metrics at each region. Validated results with permutation and null-model importance.
- Extending the framework for state-wise analysis in India for cross-country generalizability, adapting the pipeline to varying administrative geographies and sparser crowd-sourced coverage
- Presented at the **ACM IMC 2025 Student Workshop** (Madison, WI) and accepted as a poster at the **TMA 2026 PhD School** (Delft, Netherlands). Pre-print: [arXiv:2605.30809](https://arxiv.org/abs/2605.30809).

Application-Level Performance

- Investigating how application-layer performance for web browsing and video streaming tracks **Ookla** measurements, identifying to what extent speed tests are reliable predictors of user experience.
- Leading collection and analysis of YouTube and YouTube Shorts **QoE**, building an automated framework that correlates 20 streaming metrics with speed-test results under controlled network conditions to bridge network-level measurement and real-world application performance.

Web Infrastructure Security

- Studying the security, privacy, and sovereignty of U.S. state and federal web infrastructure: who can observe data in transit, where it is routed and stored, which jurisdiction governs it, and which networks operate the service.
- Developing a pipeline that traces government websites from IP to autonomous system to operating organization, producing a state-wise map of hosting strategies, security posture, and third-party vendor dependence.

Reverse-geolocating Community Anchor Institutions

- Led validation of **geographic representativeness** for **Reverse IP Geolocation (RG)**, a framework that infers IP addresses of libraries, schools, and hospitals from public street addresses, checking candidates against DNS records, WHOIS data, broadband provider data, and active measurements. Pre-print: [arXiv:2605.30791](https://arxiv.org/abs/2605.30791).

EXPERIENCE

Oracle

Bangalore, India

Senior Cloud Analyst and Consultant – *Integration & ERP Systems*

Aug 2019 – May 2022

- Worked on **Oracle Fusion Applications** to manage core business functions such as finance, procurement, and project management, developing a deep understanding of how data is stored and maintained across a large enterprise system
- Built data pipelines using **SQL** on **BI Publisher** and **Integration Cloud** that moved business records between client systems and **Oracle Cloud**.
- Translated client requirements into technical specifications end to end, and led the campus-to-corporate program, mentoring the 2020 and 2021 new-hire cohorts.

SKILLS

Languages: Python, C, Java, JavaScript, SQL

Tools: Git, BigQuery, GeoPandas, Tableau, Docker, AWS, REST & GraphQL APIs